Predicting Wildfire Ignition Cause in Canada Using Machine Learning

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Abstract

This project develops a machine learning classification model to predict whether a wildfire in Canada was naturally caused (lightning) or human-caused. Using the Canadian National Fire Database (1972–2024) containing 14,485 fires with labeled causes, combined with Environment Canada climate data, we train and evaluate multiple classifiers. Preliminary analysis achieved 91.6% accuracy and 0.891 ROC-AUC, demonstrating strong feasibility. Results will provide actionable insights for fire prevention resource allocation and targeted awareness campaigns.

1 Problem Statement

Wildfires represent one of Canada's most significant natural hazards. The 2023 fire season alone burned over 16 million hectares—more than double any previous record. Understanding wildfire *cause* is critical for prevention: while natural fires (lightning-ignited) are largely unavoidable, human-caused fires are entirely preventable through targeted interventions.

The Canadian National Fire Database (CNFDB) contains records of over 16,000 large fires (>200 hectares) from 1972–2024, with cause labels assigned by provincial fire management agencies. Approximately 86% of large fires are naturally caused, while 14% are human-caused. However, the relationship between environmental conditions and fire causation remains underexplored.

Research Question: Can we accurately predict whether a wildfire was naturally or human-caused based on temporal, geographic, and climate features? Which factors most strongly distinguish these fire types?

2 Motivation

This research addresses a practical problem with significant societal impact:

(1) **Fire Prevention**: Human-caused fires are preventable. Identifying high-risk conditions enables targeted prevention campaigns.
(2) **Resource Allocation**: Fire agencies can optimize deployment by understanding when and where different fire types occur.
(3) **Policy Development**: Evidence-based insights inform regulations on campfires, industrial activities, and land use.
(4) **Climate Context**: As fire seasons lengthen due to climate change, understanding weather-causation relationships becomes critical.

From a data science perspective, this project offers an excellent opportunity to apply classification algorithms to a real-world problem with reliable ground truth labels, interpretable features, and meaningful implications.

3 Objectives

3.1 Primary Objective

Develop a machine learning classifier that predicts wildfire ignition cause (Natural vs. Human) with accuracy exceeding 85%, using the CNFDB and Environment Canada climate data.

3.2 Secondary Objectives

- Identify key features distinguishing natural from human-caused fires through feature importance analysis
- Analyze regional and seasonal patterns in fire causation across provinces
- Integrate climate variables (temperature, precipitation, drought) to improve predictions
- Provide actionable recommendations for fire management agencies

4 Data Sources

4.1 Canadian National Fire Database

Source: Natural Resources Canada, Canadian Forest Service [1]

- **Coverage**: 1972–2024, all provinces and territories
- **Records**: 16,394 large fires; **14,485 with labeled cause**
- **Target Variable**: `CAUSE` (N=Natural, H=Human)
- **Features**: Date, province, fire size, detection method
- **Label Quality**: Assigned by professional fire investigators; 88% of records have known cause

4.2 Environment Canada Climate Data

Source: Government of Canada [2]

- **Coverage**: Daily observations from weather stations nationwide
- **Variables**: Temperature, precipitation, snow, wind speed
- **Integration**: Match fires to nearest stations for climate context

5 Methodology

5.1 Data Preparation

Load CNFDB fire data, filter to records with known cause labels, acquire climate data for fire-prone regions (BC, AB, ON, QC), and perform quality assessment.

5.2 Feature Engineering

- **Temporal**: Day of year (seasonality), month, year, weekend indicator
- **Geographic**: Province encoding
- **Climate**: Temperature anomalies, precipitation deficit
- **Fire**: Log-transformed fire size

5.3 Model Development

Train and compare: Logistic Regression (baseline), Random Forest, and Gradient Boosting. Address class imbalance (86%/14%) using class weights.

5.4 Evaluation

Stratified cross-validation with accuracy, precision, recall, F1, and ROC-AUC metrics. Feature importance analysis to interpret results.

6 Preliminary Results

A proof-of-concept using fire metadata features (no climate data yet) achieved:

Table 1: Model Performance (Test Set: 2,878 fires)

Model	Accuracy	ROC-AUC
Logistic Regression	90.8%	0.857
Random Forest	91.8%	0.888
Gradient Boosting	**91.6%**	**0.891**

Key Finding: Day of year is the strongest predictor (49% importance). Natural fires peak in July–August (lightning season), while human fires have broader seasonal distribution. Integrating climate data is expected to further improve performance.

7 Expected Contributions

(1) Validated ML model for wildfire cause prediction
(2) Analysis of temporal and regional causation patterns
(3) Recommendations for prevention resource allocation
(4) Reproducible code and methodology

8 Team Composition

- **Technical Expert** (Sai Paladugu): Machine learning model development, data pipeline, statistical analysis
- **Techincal Expert** (Ashok Sivathayalan): Problem framing, ecological interpretation, feature engineering
- **Domain Expert** (Queenie): Policy implications, literature review, visualization

References

[1] Canadian Forest Service. 2025. Canadian National Fire Database. Natural Resources Canada. `https://cwfis.cfs.nrcan.gc.ca/datamart`
[2] Environment and Climate Change Canada. Historical Climate Data. `https://dd.weather.gc.ca/`